

Smoking vs. Cancer Risk Comparison

Approximate lifetime **lung cancer** risk for people who smoke, people who don't smoke, and people who don't smoke but are exposed to secondhand smoke. Lung cancer is used because comparable percentage estimates by smoking status are strongest for this cancer. Smoking also causes many other cancers (throat, mouth, bladder, pancreas, and more).

Summary comparison

Group	Smoke ?	Secondhand smoke?	Approx. lifetime lung cancer risk	Out of 1,000 people	Takeaway
People who smoke	Yes	N/A (own smoking dominates)	~15% (range ~11–22%)	~150	Much higher risk; rises with more years and cigarettes smoked. About 15–25× higher than never-smokers.
People who do not smoke (little/no secondhand smoke)	No	No / low	~1.5% (range ~1–2%)	~15	Much lower chance. Lung cancer can still occur from radon, pollution, genetics, etc.
People who do not smoke but get secondhand smoke	No	Yes (home or work)	~1.9% (~20–30% higher than baseline)	~19	Higher than unexposed never-smokers, but far below active smokers. ~7,300 U.S. lung cancer deaths/year from SHS among nonsmokers.

At-a-glance

Category	Approx. % (lifetime lung cancer)	Compared with never-smokers (no SHS)
Smoke cigarettes	~15%	About 15–25 times higher
Do not smoke (no secondhand smoke)	~1.5%	Baseline
Do not smoke + secondhand smoke	~1.9%	About 20–30% higher (relative)

How these numbers were chosen

Smokers ~15%: Midpoint near commonly reported current-smoker lifetime risks (~11–22% depending on sex, study, and smoking intensity).

Never-smokers ~1.5%: Midpoint of roughly 1–2% lifetime/incidence-style estimates from large cohort summaries.

Secondhand-only ~1.9%: Apply about a 25% relative increase (middle of the U.S. Surgeon General's 20–30% increase / meta-analysis $RR \approx 1.24$) to the 1.5% baseline $\rightarrow 1.5\% \times 1.25 \approx 1.9\%$. A Japanese life-table study similarly found higher absolute lifetime mortality for never-smokers with SHS than without (men 3.2%→4.1%; women 1.9%→2.4%).

Key caveats

- These are **population averages**, not a prediction for one person. Dose, duration, age started smoking, quitting, sex, radon, asbestos, and genetics all matter.

- Absolute secondhand-smoke risk stays much closer to never-smokers than to smokers, but SHS still causes thousands of U.S. lung cancer deaths among nonsmokers each year.
- CDC: people who smoke are about **15–30 times** more likely to get or die from lung cancer than people who do not smoke.

Selected sources

- CDC — Lung Cancer Risk Factors: <https://www.cdc.gov/lung-cancer/risk-factors/index.html>
- CDC — Lung Cancer Among People Who Never Smoked: <https://www.cdc.gov/lung-cancer/nonsmokers/index.html>
- NCI — Secondhand Smoke and Cancer:
<https://www.cancer.gov/about-cancer/causes-prevention/risk/tobacco/second-hand-smoke-fact-sheet>
- U.S. Surgeon General (involuntary smoking summary): <https://www.ncbi.nlm.nih.gov/books/NBK44330/>
- American Cancer Society — Lung Cancer Key Statistics: <https://www.cancer.org/cancer/types/lung-cancer/about/key-statistics.html>
- Thun et al., PLOS Medicine 2008 (never-smoker cumulative risk):
<https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.0050185>
- Inoue et al., JCO 2020 (Japan lifetime risk + SHS): <https://doi.org/10.1093/jco/hyaa094>
- ERS Review 2024 meta-analysis (SHS RR $\approx$ 1.24): <https://publications.ersnet.org/content/errev/33/174/240077>

Disclaimer: This PDF summarizes published estimates for educational comparison. It is not medical advice. Talk with a clinician for personal risk assessment. Quitting smoking and avoiding secondhand smoke lower cancer risk.